



Performance Analysis

in GPU Programming

Presenting

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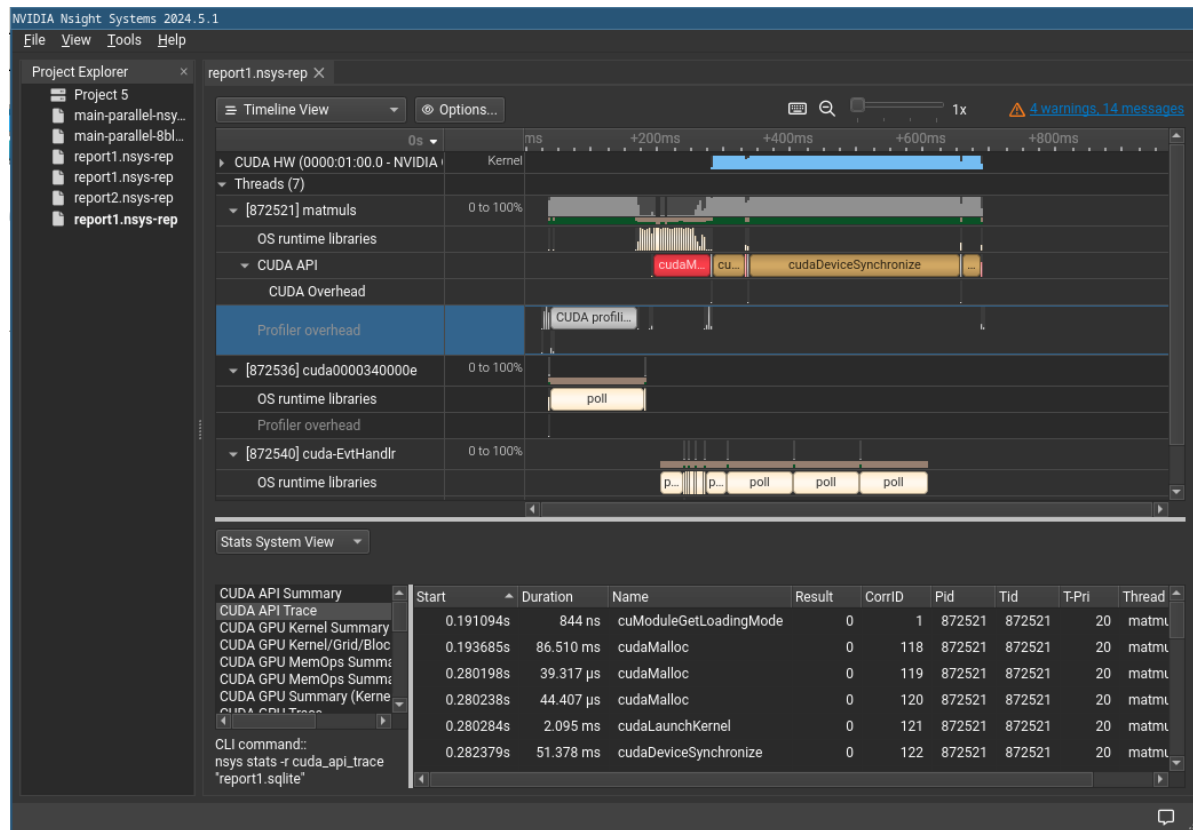
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Profiling

- *Performance analysis* is the process of studying how a program behaves and performs, using information collected about the program. (for example, using a profiling tool).
- *Profiling* is the process of measuring a program's behaviour during execution (execution time, memory usage, latency etc.)
- *Metrics* are measurable quantities that shows a particular aspect of a program.
- NVIDIA provides two profiling tools with CUDA toolkit.
 - Nsight Systems
 - Nsight Compute

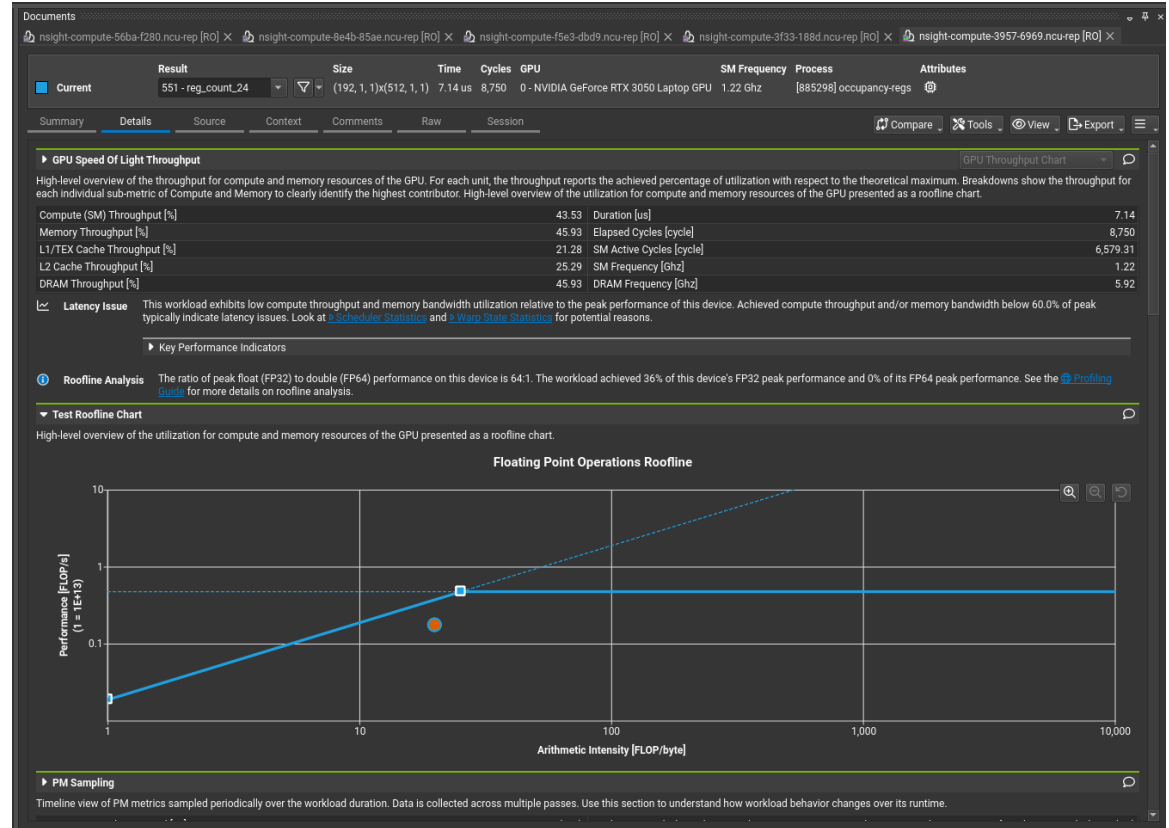
Nsight Systems

- *Nsight systems* is a profiling tool for GPU programs.
 - Shows CPU-GPU interaction
 - Show parallel streams
 - Shows CPU threads
 - Durations of API calls and kernels.



Nsight Compute

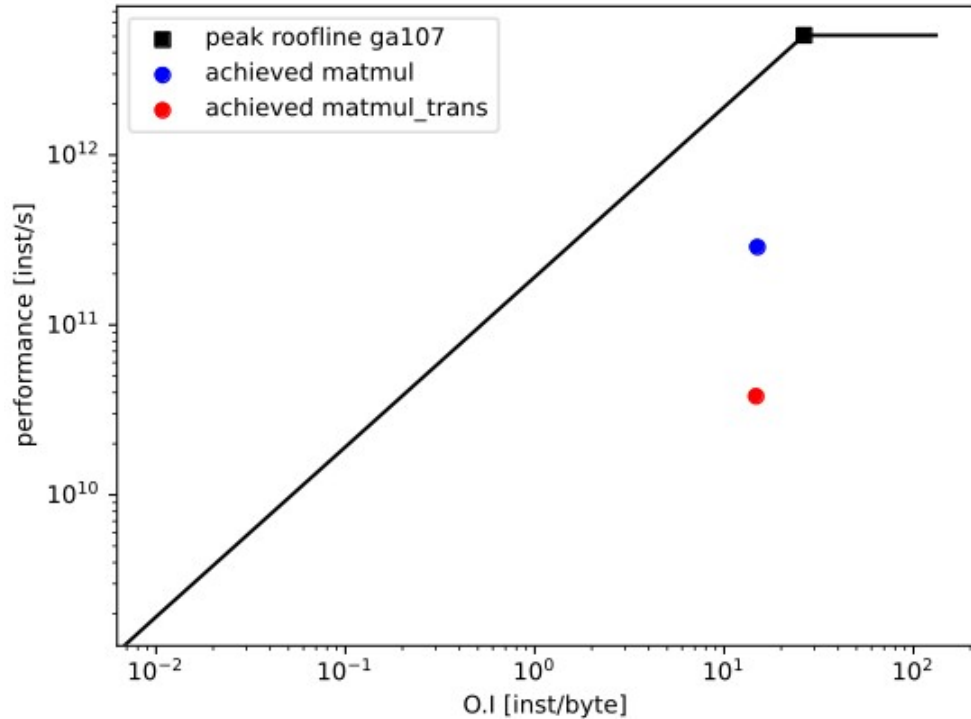
- *Nsight compute* is a profiling tool for specifically for GPU kernels.
- Provides detailed information about kernels.
- Information about SMs, caches, and device memory.
- Does not show individual streams or threads.





Let's have a look at Nsight Compute and Nsight Systems

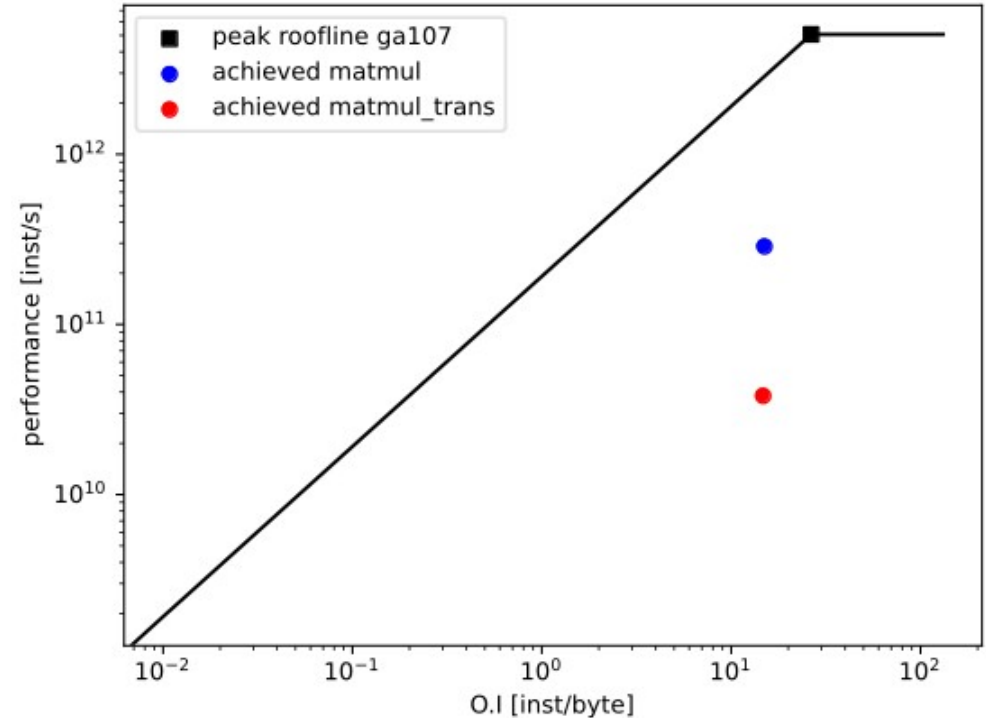
Roofline Example



- Roofline used to compare two kernels.
 - One is regular matrix multiplication
 - Other accesses elements in transposed order
- The transposed order shows lower performance due to having uncoalesced access, slowing down the kernel.
- Arithmetic intensity remains the same because the work size is the same and no. of instructions are the same.

Roofline Model

- Roofline model is a performance model used used to visualize the performance of a kernel on a system.
- Kernels can be compared using this model and we can obtain certain insights about the kernel's performance.
 - Compute bound or memory bound
 - Does the kernel achieve maximum possible performance.



Command Line Arguments

- Usage: `ncu [ncu options] [program] [program args]`
- Options

<code>--export</code>	Exports a report with filename <code><name>.ncu-rep</code>
<code>--import <file></code>	Imports an existing report
<code>--force-overwrite</code>	Overwrites an already existing report
<code>--metrics <m1,m2,...></code>	Comma separated list of metrics to be collected
<code>--set <set-id></code> <code>--list-set</code>	Collects metrics in the set <code><set-id></code>
<code>--query-metrics</code>	Prints out available metrics for this GPU
<code>--open-in-ui</code>	Directly opens in UI after profiling a program
<code>--help</code>	Shows all available options (check this for more available options)



Thank You